10 minutes to pandas
Intro to data structures
Essential basic functionality
IO tools (text, CSV, HDF5, …)
PyArrow Functionality
Indexing and selecting data
MultiIndex / advanced indexing
Copy-on-Write (CoW)
Merge, join, concatenate and compare
Reshaping and pivot tables
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# Merge, join, concatenate and compare

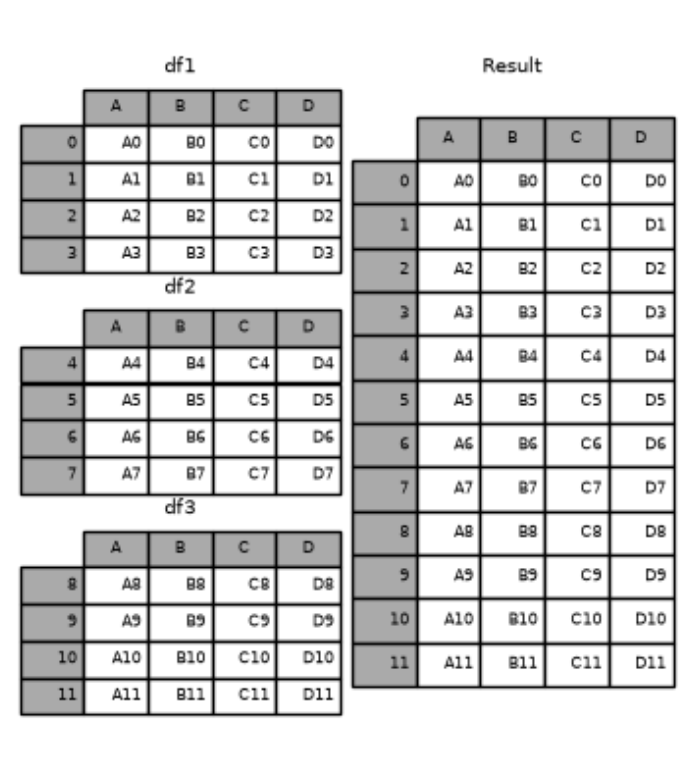
pandas provides various methods for combining and comparing `Series` or `DataFrame`.

- `concat()`: Merge multiple `Series` or `DataFrame` objects along a shared index or column
- `DataFrame.join()`: Merge multiple `DataFrame` objects along the columns
- `DataFrame.combine_first()`: Update missing values with non-missing values in the same location
- `merge()`: Combine two `Series` or `DataFrame` objects with SQL-style joining
- `merge_ordered()`: Combine two `Series` or `DataFrame` objects along an ordered axis
- `merge_asof()`: Combine two `Series` or `DataFrame` objects by near instead of exact matching keys
- `Series.compare()` and `DataFrame.compare()`: Show differences in values between two `Series` or `DataFrame` objects

## `concat()`

The `concat()` function concatenates an arbitrary amount of `Series` or `DataFrame` objects along an axis while performing optional set logic (union or intersection) of the indexes on the other axes. Like `numpy.concatenate`, `concat()` takes a list or dict of homogeneously-typed objects and concatenates them.

```
In [1]: df1 = pd.DataFrame(
   ...:     {
   ...:         "A": ["A0", "A1", "A2", "A3"],
   ...:         "B": ["B0", "B1", "B2", "B3"],
   ...:         "C": ["C0", "C1", "C2", "C3"],
   ...:         "D": ["D0", "D1", "D2", "D3"],
   ...:     },
   ...:     index=[0, 1, 2, 3],
   ...: )
   ...: 

In [2]: df2 = pd.DataFrame(
   ...:     {
   ...:         "A": ["A4", "A5", "A6", "A7"],
   ...:         "B": ["B4", "B5", "B6", "B7"],
   ...:         "C": ["C4", "C5", "C6", "C7"],
   ...:         "D": ["D4", "D5", "D6", "D7"],
   ...:     },
   ...:     index=[4, 5, 6, 7],
   ...: )
   ...: 

In [3]: df3 = pd.DataFrame(
   ...:     {
   ...:         "A": ["A8", "A9", "A10", "A11"],
   ...:         "B": ["B8", "B9", "B10", "B11"],
   ...:         "C": ["C8", "C9", "C10", "C11"],
   ...:         "D": ["D8", "D9", "D10", "D11"],
   ...:     },
   ...:     index=[8, 9, 10, 11],
   ...: )
   ...: 

In [4]: frames = [df1, df2, df3]

In [5]: result = pd.concat(frames)

In [6]: result
Out[6]: 
      A    B    C    D
0    A0   B0   C0   D0
1    A1   B1   C1   D1
2    A2   B2   C2   D2
3    A3   B3   C3   D3
4    A4   B4   C4   D4
5    A5   B5   C5   D5
6    A6   B6   C6   D6
7    A7   B7   C7   D7
8    A8   B8   C8   D8
9    A9   B9   C9   D9
10  A10  B10  C10  D10
11  A11  B11  C11  D11
```



**Note**

`concat()` makes a full copy of the data, and iteratively reusing `concat()` can create unnecessary copies. Collect all `DataFrame` or `Series` objects in a list before using `concat()`.

```
frames = [process_your_file(f) for f in files]
result = pd.concat(frames)
```

**Note**

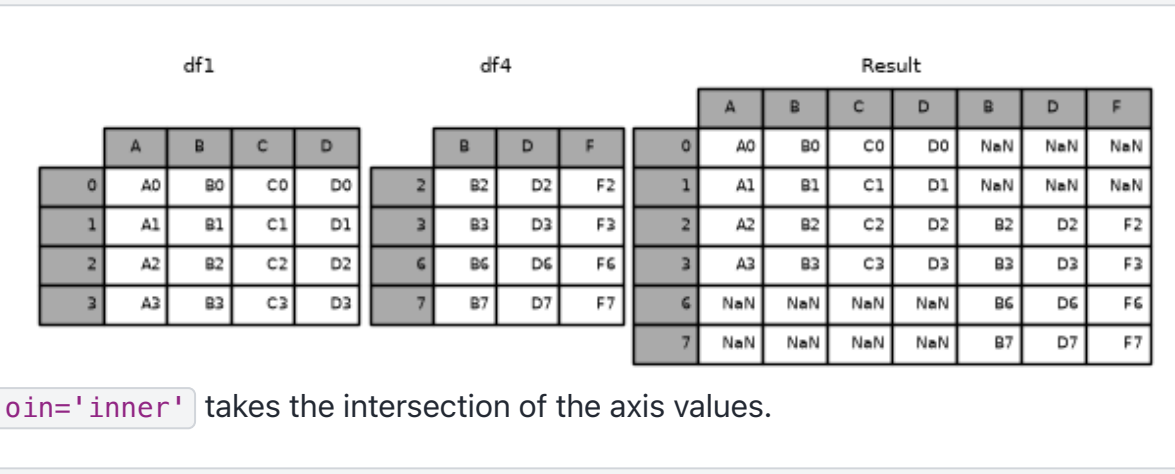
When concatenating `DataFrame` with named axes, pandas will attempt to preserve these index/column names whenever possible. In the case where all inputs share a common name, this name will be assigned to the result. When the input names do not all agree, the result will be unnamed. The same is true for `MultiIndex`, but the logic is applied separately on a level-by-level basis.

### Joining logic of the resulting axis

The `join` keyword specifies how to handle axis values that don't exist in the first `DataFrame`.

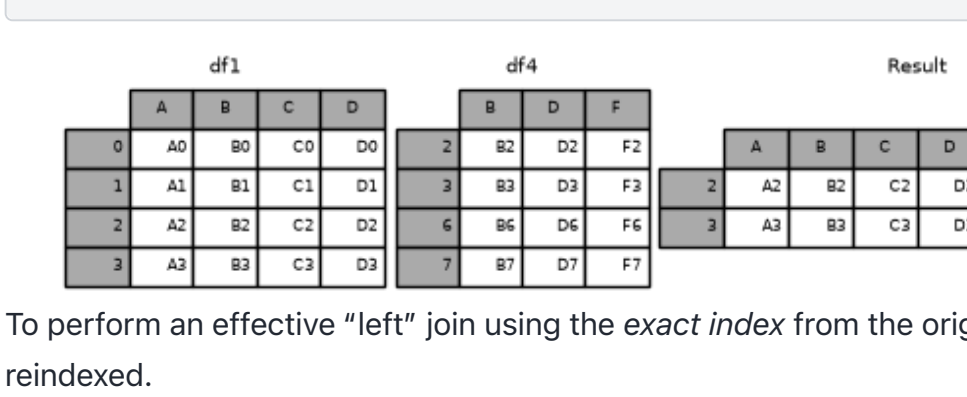
`join='outer'` takes the union of all axis values.

```
In [7]: df4 = pd.DataFrame(
   ...:     {
   ...:         "B": ["B2", "B3", "B6", "B7"],
   ...:         "D": ["D2", "D3", "D6", "D7"],
   ...:         "F": ["F2", "F3", "F6", "F7"],
   ...:     },
   ...:     index=[2, 3, 6, 7],
   ...: )
   ...: 

In [8]: result = pd.concat([df1, df4], axis=1)

In [9]: result
Out[9]: 
     A    B    C    D    B    D    F
0   A0   B0   C0   D0  NaN  NaN  NaN
1   A1   B1   C1   D1  NaN  NaN  NaN
2   A2   B2   C2   D2   B2   D2   F2
3   A3   B3   C3   D3   B3   D3   F3
6  NaN  NaN  NaN  NaN   B6   D6   F6
7  NaN  NaN  NaN  NaN   B7   D7   F7
```



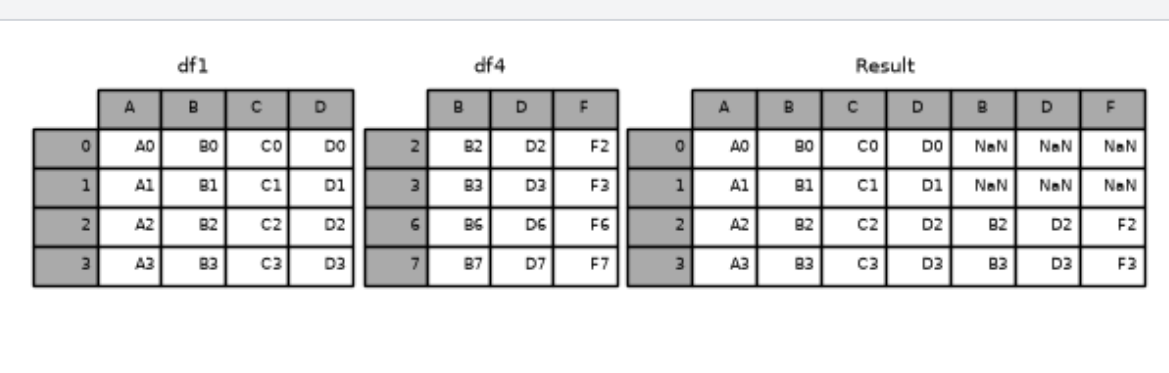
`join='inner'` takes the intersection of the axis values.

```
In [10]: result = pd.concat([df1, df4], axis=1, join="inner")

In [11]: result
Out[11]: 
    A   B   C   D   B   D   F
2  A2  B2  C2  D2  B2  D2  F2
3  A3  B3  C3  D3  B3  D3  F3
```



To perform an effective "left" join using the *exact index* from the original `DataFrame`, result can be reindexed.

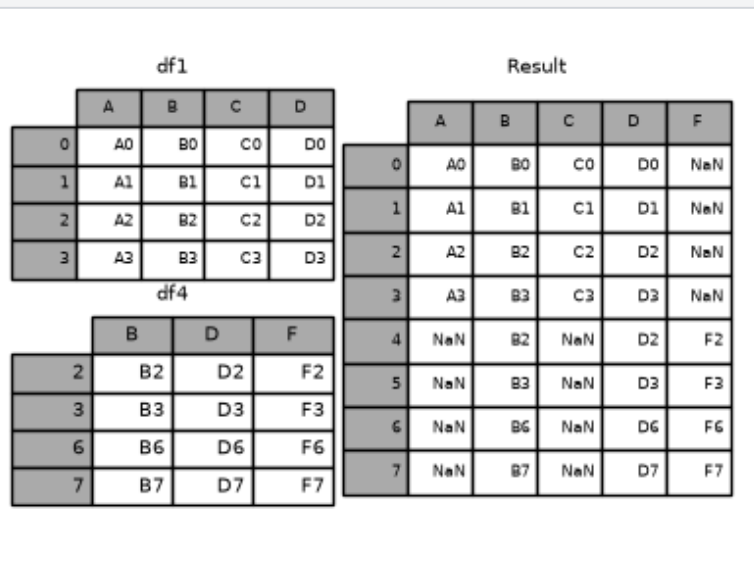
```
In [12]: result = pd.concat([df1, df4], axis=1).reindex(df1.index)

In [13]: result
Out[13]: 
    A   B   C   D    B    D    F
0  A0  B0  C0  D0  NaN  NaN  NaN
1  A1  B1  C1  D1  NaN  NaN  NaN
2  A2  B2  C2  D2   B2   D2   F2
3  A3  B3  C3  D3   B3   D3   F3
```



### Ignoring indexes on the concatenation axis

For `DataFrame` objects which don't have a meaningful index, the `ignore_index` ignores overlapping indexes.

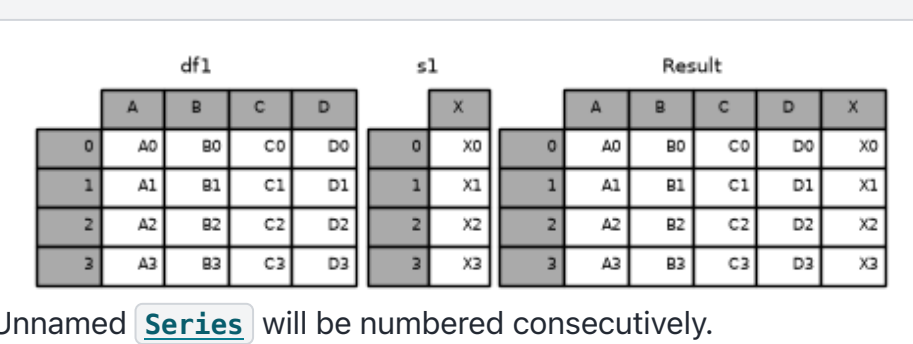
```
In [14]: result = pd.concat([df1, df4], ignore_index=True, sort=False)

In [15]: result
Out[15]: 
     A   B    C   D    F
0   A0  B0   C0  D0  NaN
1   A1  B1   C1  D1  NaN
2   A2  B2   C2  D2  NaN
3   A3  B3   C3  D3  NaN
4  NaN  B2  NaN  D2   F2
5  NaN  B3  NaN  D3   F3
6  NaN  B6  NaN  D6   F6
7  NaN  B7  NaN  D7   F7
```



### Concatenating `Series` and `DataFrame` together

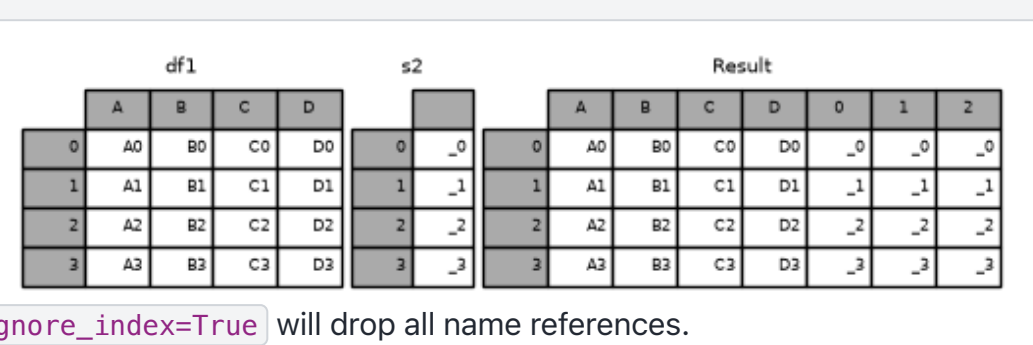
You can concatenate a mix of `Series` and `DataFrame` objects. The `Series` will be transformed to `DataFrame` with the column name as the name of the `Series`.

```
In [16]: s1 = pd.Series(["X0", "X1", "X2", "X3"], name="X")

In [17]: result = pd.concat([df1, s1], axis=1)

In [18]: result
Out[18]: 
    A   B   C   D   X
0  A0  B0  C0  D0  X0
1  A1  B1  C1  D1  X1
2  A2  B2  C2  D2  X2
3  A3  B3  C3  D3  X3
```



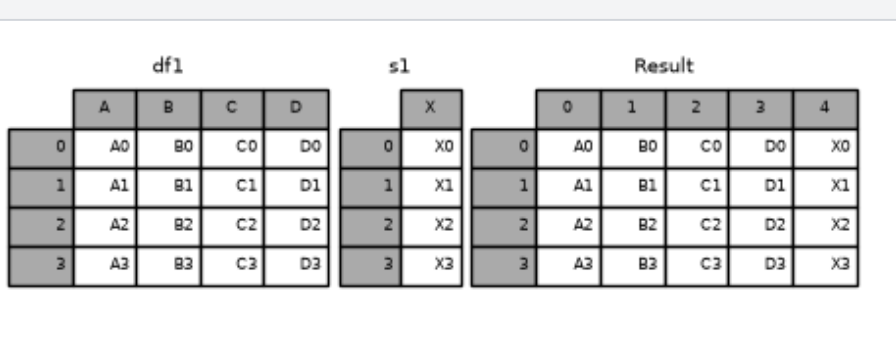
Unnamed `Series` will be numbered consecutively.

```
In [19]: s2 = pd.Series(["_0", "_1", "_2", "_3"])

In [20]: result = pd.concat([df1, s2, s2, s2], axis=1)

In [21]: result
Out[21]: 
    A   B   C   D   0   1   2
0  A0  B0  C0  D0  _0  _0  _0
1  A1  B1  C1  D1  _1  _1  _1
2  A2  B2  C2  D2  _2  _2  _2
3  A3  B3  C3  D3  _3  _3  _3
```



`ignore_index=True` will drop all name references.

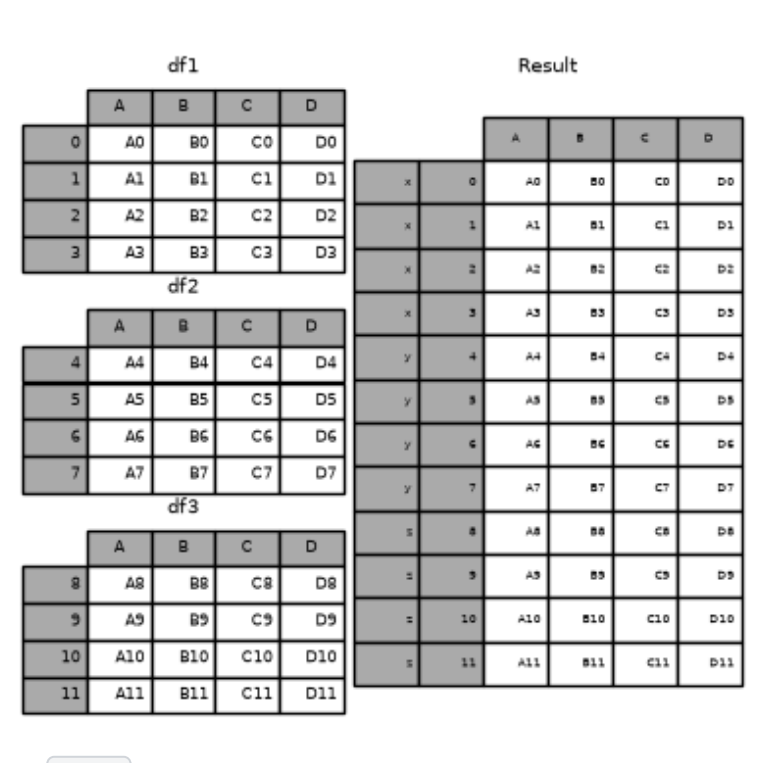
```
In [22]: result = pd.concat([df1, s1], axis=1, ignore_index=True)

In [23]: result
Out[23]: 
    0   1   2   3   4
0  A0  B0  C0  D0  X0
1  A1  B1  C1  D1  X1
2  A2  B2  C2  D2  X2
3  A3  B3  C3  D3  X3
```



### Resulting `keys`

The `keys` argument adds another axis level to the resulting index or column (creating a `MultiIndex`) associate specific keys with each original `DataFrame`.

```
In [24]: result = pd.concat(frames, keys=["x", "y", "z"])

In [25]: result
Out[25]: 
        A    B    C    D
x 0    A0   B0   C0   D0
  1    A1   B1   C1   D1
  2    A2   B2   C2   D2
  3    A3   B3   C3   D3
y 4    A4   B4   C4   D4
  5    A5   B5   C5   D5
  6    A6   B6   C6   D6
  7    A7   B7   C7   D7
z 8    A8   B8   C8   D8
  9    A9   B9   C9   D9
  10  A10  B10  C10  D10
  11  A11  B11  C11  D11

In [26]: result.loc["y"]
Out[26]: 
    A   B   C   D
4  A4  B4  C4  D4
5  A5  B5  C5  D5
6  A6  B6  C6  D6
7  A7  B7  C7  D7
```

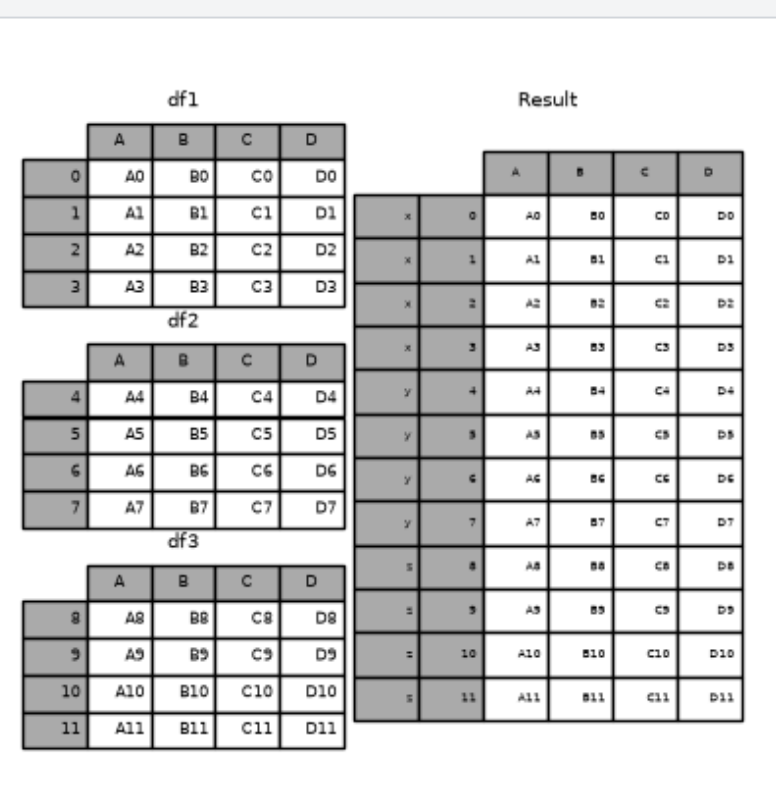


The `keys` argument can override the column names when creating a new `DataFrame` based on existing `Series`.

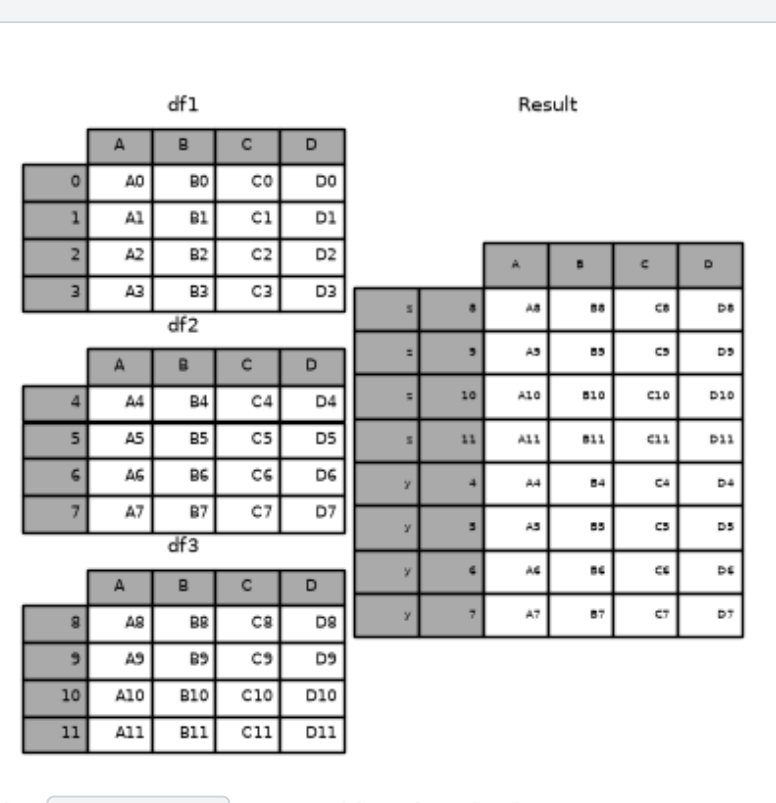
```
In [27]: s3 = pd.Series([0, 1, 2, 3], name="foo")

In [28]: s4 = pd.Series([0, 1, 2, 3])

In [29]: s5 = pd.Series([0, 1, 4, 5])

In [30]: pd.concat([s3, s4, s5], axis=1)
Out[30]: 
   foo  0  1
0    0  0  0
1    1  1  1
2    2  2  4
3    3  3  5

In [31]: pd.concat([s3, s4, s5], axis=1, keys=["red", "blue", "yellow"])
Out[31]: 
   red  blue  yellow
0    0     0       0
1    1     1       1
2    2     2       4
3    3     3       5
```

You can also pass a dict to `concat()` in which case the dict keys will be used for the `keys` argument unless other `keys` argument is specified:

```
In [32]: pieces = {"x": df1, "y": df2, "z": df3}

In [33]: result = pd.concat(pieces)

In [34]: result
Out[34]: 
        A    B    C    D
x 0    A0   B0   C0   D0
  1    A1   B1   C1   D1
  2    A2   B2   C2   D2
  3    A3   B3   C3   D3
y 4    A4   B4   C4   D4
  5    A5   B5   C5   D5
  6    A6   B6   C6   D6
  7    A7   B7   C7   D7
z 8    A8   B8   C8   D8
  9    A9   B9   C9   D9
  10  A10  B10  C10  D10
  11  A11  B11  C11  D11
```



```
In [35]: result = pd.concat(pieces, keys=["z", "y"])

In [36]: result
Out[36]: 
        A    B    C    D
z 8    A8   B8   C8   D8
  9    A9   B9   C9   D9
  10  A10  B10  C10  D10
  11  A11  B11  C11  D11
y 4    A4   B4   C4   D4
  5    A5   B5   C5   D5
  6    A6   B6   C6   D6
  7    A7   B7   C7   D7
```

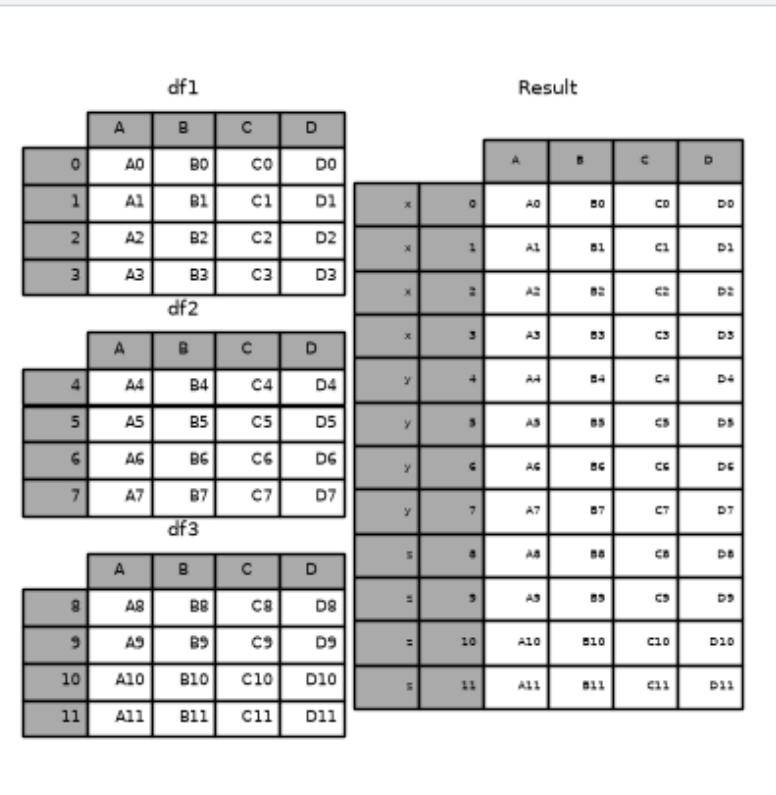


The `MultiIndex` created has levels that are constructed from the passed keys and the index of the `DataFrame` pieces:

```
In [37]: result.index.levels
Out[37]: FrozenList([['z', 'y'], [4, 5, 6, 7, 8, 9, 10, 11]])
```

`levels` argument allows specifying resulting levels associated with the `keys`.

```
In [38]: result = pd.concat(
   ....:     pieces, keys=["x", "y", "z"], levels=[["z", "y", "x", "w"]], names=["g
   ....: )
   ....: 

In [39]: result
Out[39]: 
                A    B    C    D
group_key                      
x         0    A0   B0   C0   D0
          1    A1   B1   C1   D1
          2    A2   B2   C2   D2
          3    A3   B3   C3   D3
y         4    A4   B4   C4   D4
          5    A5   B5   C5   D5
          6    A6   B6   C6   D6
          7    A7   B7   C7   D7
z         8    A8   B8   C8   D8
          9    A9   B9   C9   D9
          10  A10  B10  C10  D10
          11  A11  B11  C11  D11
```

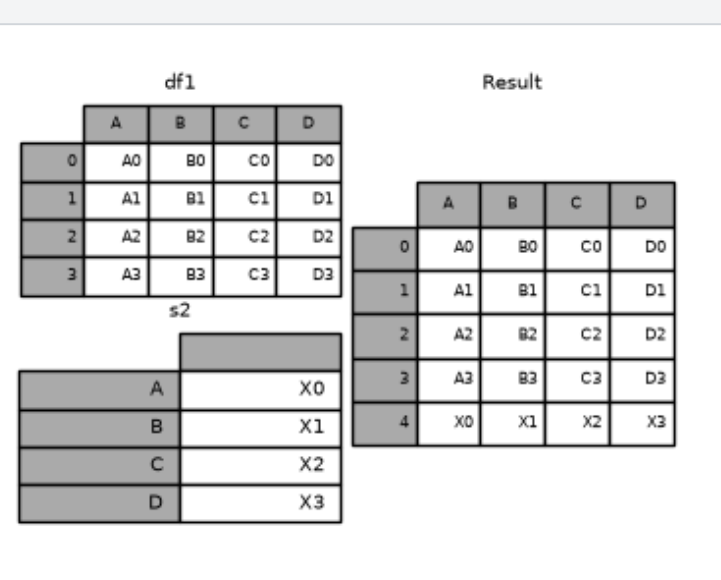


```
In [40]: result.index.levels
Out[40]: FrozenList([['z', 'y', 'x', 'w'], [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]])
```

### Appending rows to a `DataFrame`

If you have a `Series` that you want to append as a single row to a `DataFrame`, you can convert the row into a `DataFrame` and use `concat()`.

```
In [41]: s2 = pd.Series(["X0", "X1", "X2", "X3"], index=["A", "B", "C", "D"])

In [42]: result = pd.concat([df1, s2.to_frame().T], ignore_index=True)

In [43]: result
Out[43]: 
    A   B   C   D
0  A0  B0  C0  D0
1  A1  B1  C1  D1
2  A2  B2  C2  D2
3  A3  B3  C3  D3
4  X0  X1  X2  X3
```



## `merge()`

`merge()` performs join operations similar to relational databases like SQL. Users who are familiar with SQL but new to pandas can reference a comparison with SQL.

### Merge types

`merge()` implements common SQL style joining operations.

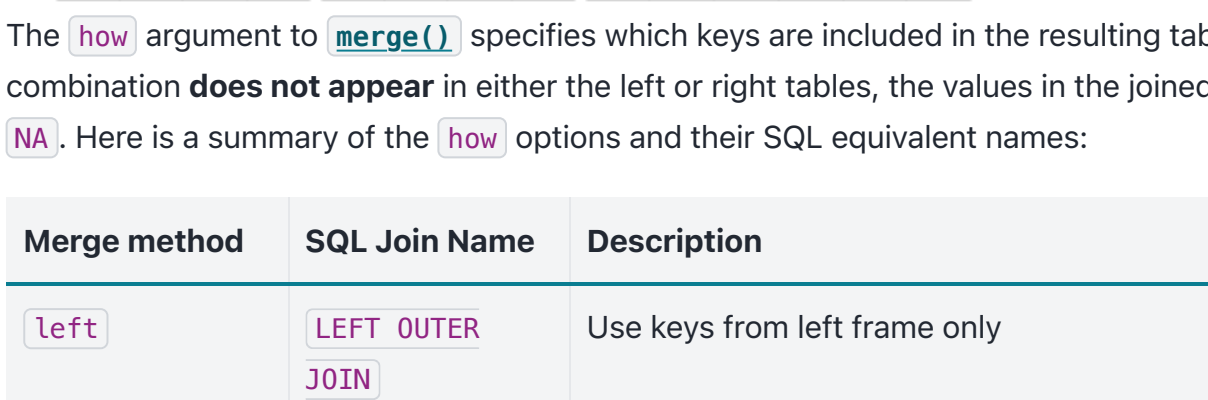
- **one-to-one**: joining two `DataFrame` objects on their indexes which must contain unique values.
- **many-to-one**: joining a unique index to one or more columns in a different `DataFrame`.
- **many-to-many**: joining columns on columns.

**Note**

When joining columns on columns, potentially a many-to-many join, any indexes on the passed `DataFrame` objects **will be discarded**.

For a **many-to-many** join, if a key combination appears more than once in both tables, the `DataFrame` will have the **Cartesian product** of the associated data.

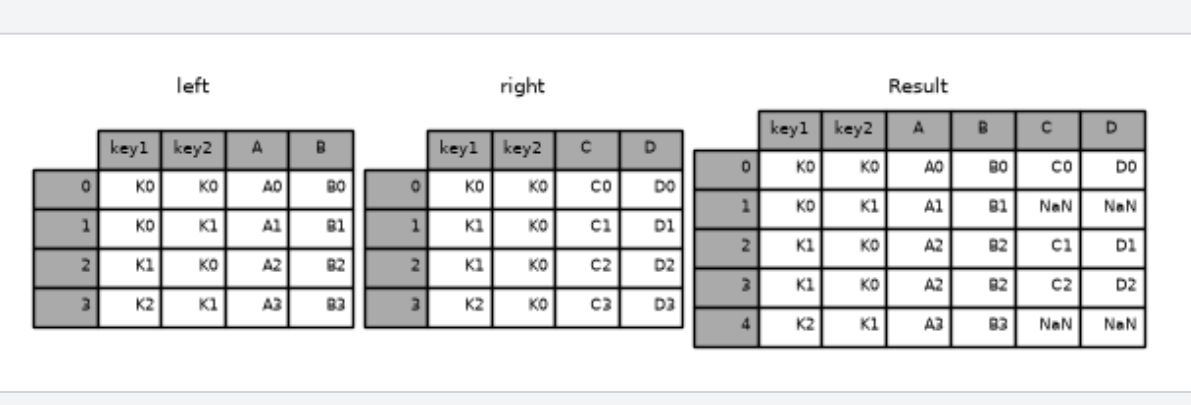
```
In [44]: left = pd.DataFrame(
   ....:     {
   ....:         "key": ["K0", "K1", "K2", "K3"],
   ....:         "A": ["A0", "A1", "A2", "A3"],
   ....:         "B": ["B0", "B1", "B2", "B3"],
   ....:     }
   ....: )
   ....: 

In [45]: right = pd.DataFrame(
   ....:     {
   ....:         "key": ["K0", "K1", "K2", "K3"],
   ....:         "C": ["C0", "C1", "C2", "C3"],
   ....:         "D": ["D0", "D1", "D2", "D3"],
   ....:     }
   ....: )
   ....: 

In [46]: result = pd.merge(left, right, on="key")

In [47]: result
Out[47]: 
  key   A   B   C   D
0  K0  A0  B0  C0  D0
1  K1  A1  B1  C1  D1
2  K2  A2  B2  C2  D2
3  K3  A3  B3  C3  D3
```



The `how` argument to `merge()` specifies which keys are included in the resulting table. If a key combination **does not appear** in either the left or right tables, the values in the joined table will be `NA`. Here is a summary of the `how` options and their SQL equivalent names:

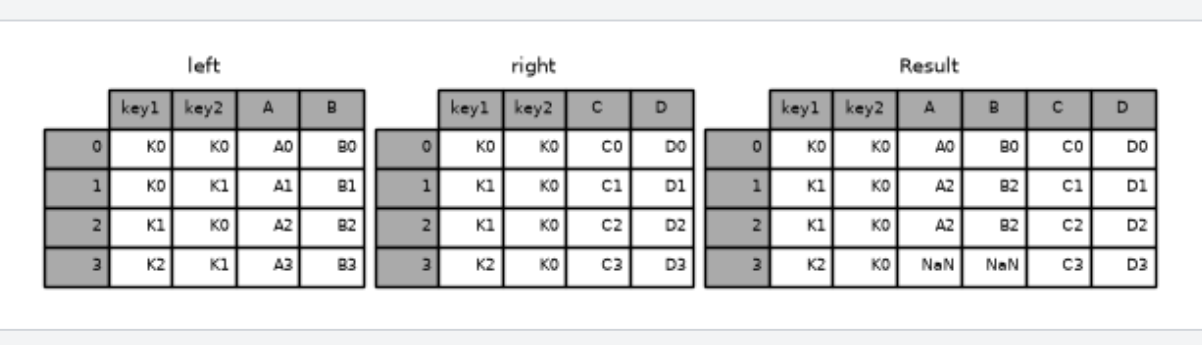
| Merge method | SQL Join Name | Description |
|---|---|---|
| `left` | `LEFT OUTER JOIN` | Use keys from left frame only |
| `right` | `RIGHT OUTER JOIN` | Use keys from right frame only |
| `outer` | `FULL OUTER JOIN` | Use union of keys from both frames |
| `inner` | `INNER JOIN` | Use intersection of keys from both frames |
| `cross` | `CROSS JOIN` | Create the cartesian product of rows of both frames |

```
In [48]: left = pd.DataFrame(
   ....:     {
   ....:         "key1": ["K0", "K0", "K1", "K2"],
   ....:         "key2": ["K0", "K1", "K0", "K1"],
   ....:         "A": ["A0", "A1", "A2", "A3"],
   ....:         "B": ["B0", "B1", "B2", "B3"],
   ....:     }
   ....: )
   ....: 

In [49]: right = pd.DataFrame(
   ....:     {
   ....:         "key1": ["K0", "K1", "K1", "K2"],
   ....:         "key2": ["K0", "K0", "K0", "K0"],
   ....:         "C": ["C0", "C1", "C2", "C3"],
   ....:         "D": ["D0", "D1", "D2", "D3"],
   ....:     }
   ....: )
   ....: 

In [50]: result = pd.merge(left, right, how="left", on=["key1", "key2"])

In [51]: result
Out[51]: 
  key1 key2   A   B    C    D
0   K0   K0  A0  B0   C0   D0
1   K0   K1  A1  B1  NaN  NaN
2   K1   K0  A2  B2   C1   D1
3   K1   K0  A2  B2   C2   D2
4   K2   K1  A3  B3  NaN  NaN
```

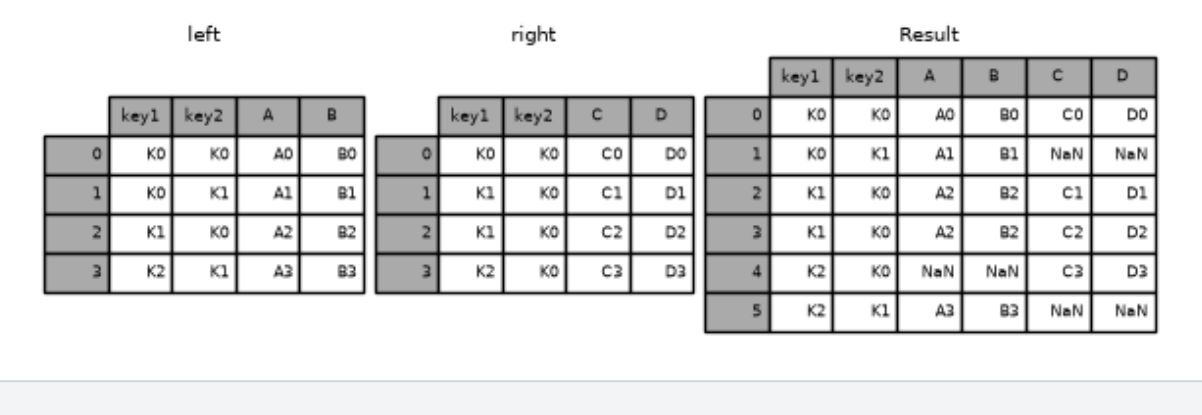


```
In [52]: result = pd.merge(left, right, how="right", on=["key1", "key2"])

In [53]: result
Out[53]: 
  key1 key2   A    B   C   D
```

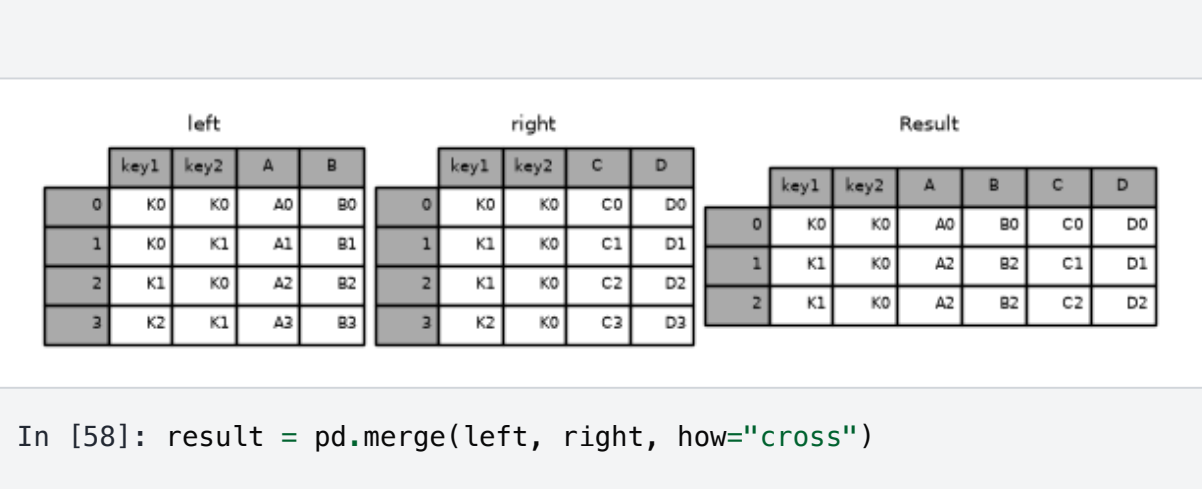
```
0   K0   K0   A0   B0  C0  D0
1   K1   K0   A2   B2  C1  D1
2   K1   K0   A2   B2  C2  D2
3   K2   K0  NaN  NaN  C3  D3
```



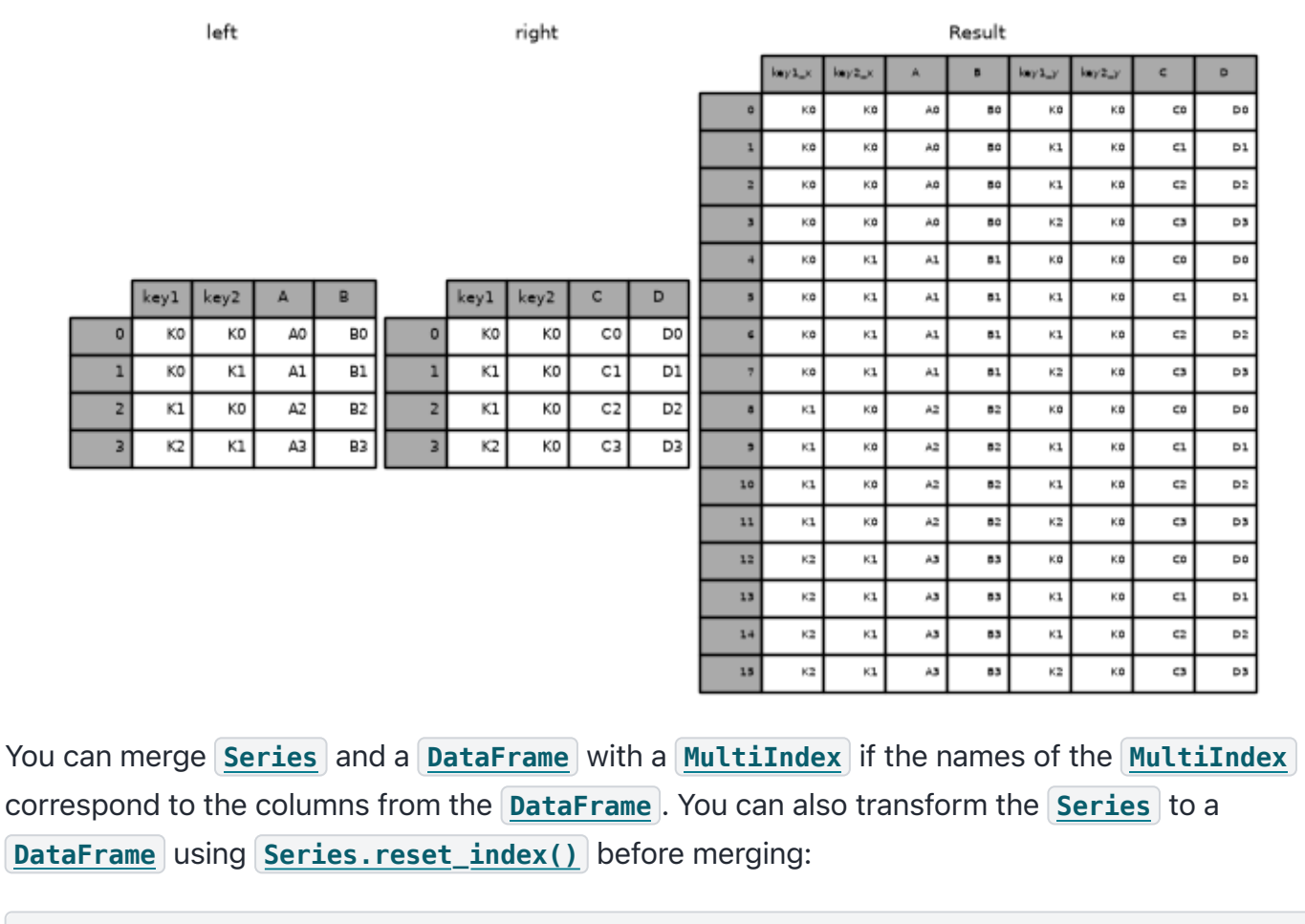
```
In [54]: result = pd.merge(left, right, how="outer", on=["key1", "key2"])

In [55]: result
Out[55]: 
  key1 key2    A    B    C    D
0   K0   K0   A0   B0   C0   D0
1   K0   K1   A1   B1  NaN  NaN
2   K1   K0   A2   B2   C1   D1
3   K1   K0   A2   B2   C2   D2
4   K2   K0  NaN  NaN   C3   D3
5   K2   K1   A3   B3  NaN  NaN
```



```
In [56]: result = pd.merge(left, right, how="inner", on=["key1", "key2"])

In [57]: result
Out[57]: 
  key1 key2   A   B   C   D
0   K0   K0  A0  B0  C0  D0
1   K1   K0  A2  B2  C1  D1
2   K1   K0  A2  B2  C2  D2
```



```
In [58]: result = pd.merge(left, right, how="cross")

In [59]: result
Out[59]: 
   key1_x key2_x   A   B key1_y key2_y   C   D
0      K0     K0  A0  B0     K0     K0  C0  D0
1      K0     K0  A0  B0     K1     K0  C1  D1
2      K0     K0  A0  B0     K1     K0  C2  D2
3      K0     K0  A0  B0     K2     K0  C3  D3
4      K0     K1  A1  B1     K0     K0  C0  D0
..    ...    ...  ..  ..    ...    ...  ..  ..
11     K1     K0  A2  B2     K2     K0  C3  D3
12     K2     K1  A3  B3     K0     K0  C0  D0
13     K2     K1  A3  B3     K1     K0  C1  D1
14     K2     K1  A3  B3     K1     K0  C2  D2
15     K2     K1  A3  B3     K2     K0  C3  D3

[16 rows x 8 columns]
```

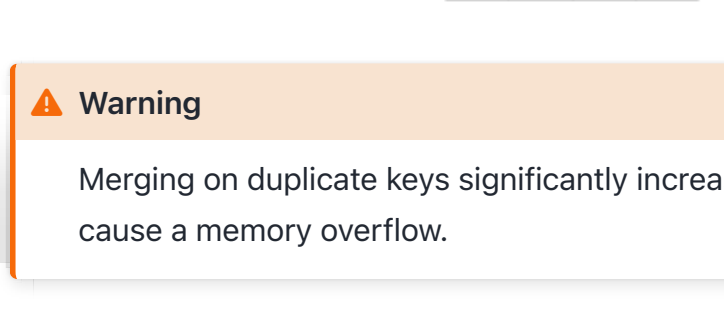


You can merge `Series` and a `DataFrame` with a `MultiIndex` if the names of the `MultiIndex` correspond to the columns from the `DataFrame`. You can also transform the `Series` to a `DataFrame` using `Series.reset_index()` before merging:

```
In [60]: df = pd.DataFrame({"Let": ["A", "B", "C"], "Num": [1, 2, 3]})

In [61]: df
Out[61]: 
  Let  Num
0   A    1
1   B    2
2   C    3

In [62]: ser = pd.Series(
   ....:     ["a", "b", "c", "d", "e", "f"],
   ....:     index=pd.MultiIndex.from_arrays(
   ....:         [["A", "B", "C"] * 2, [1, 2, 3, 4, 5, 6]], names=["Let", "Num"]
   ....:     ),
   ....: )
   ....: 

In [63]: ser
Out[63]: 
Let  Num
A    1      a
B    2      b
C    3      c
A    4      d
B    5      e
C    6      f
dtype: str

In [64]: pd.merge(df, ser.reset_index(), on=["Let", "Num"])
Out[64]: 
  Let  Num  0
0   A    1  a
1   B    2  b
2   C    3  c
```

Performing an outer join with duplicate join keys in `DataFrame`:

```
In [65]: left = pd.DataFrame({"A": [1, 2], "B": [2, 2]})

In [66]: right = pd.DataFrame({"A": [4, 5, 6], "B": [2, 2, 2]})

In [67]: result = pd.merge(left, right, on="B", how="outer")

In [68]: result
Out[68]: 
   A_x  B  A_y
0    1  2    4
1    1  2    5
2    1  2    6
3    2  2    4
4    2  2    5
5    2  2    6
```



> **Warning**
>
> Merging on duplicate keys significantly increase the dimensions of the result and can cause a memory overflow.

## Merge key uniqueness

The `validate` argument checks whether the uniqueness of merge keys. Key uniqueness is checked before merge operations and can protect against memory overflows and unexpected key duplication.

```
In [69]: left = pd.DataFrame({"A": [1, 2], "B": [1, 2]})

In [70]: right = pd.DataFrame({"A": [4, 5, 6], "B": [2, 2, 2]})

In [71]: result = pd.merge(left, right, on="B", how="outer", validate="one_to_one")
---------------------------------------------------------------------------
MergeError                                Traceback (most recent call last)
Cell In[71], line 1
----> 1 result = pd.merge(left, right, on="B", how="outer", validate="one_to_one")

File ~/work/pandas/pandas/pandas/core/reshape/merge.py:385, in merge(left, right, ho
    371     return _cross_merge(
    372         left_df,
    373         right_df,
   (...)    382         validate=validate,
    383     )
    384 else:
--> 385     op = _MergeOperation(
    386         left_df,
    387         right_df,
    388         how=how,
    389         on=on,
    390         left_on=left_on,
    391         right_on=right_on,
    392         left_index=left_index,
    393         right_index=right_index,
    394         sort=sort,
    395         suffixes=suffixes,
    396         indicator=indicator,
    397         validate=validate,
    398     )
    399     return op.get_result()

File ~/work/pandas/pandas/pandas/core/reshape/merge.py:1037, in _MergeOperation.__i
   1033 # If argument passed to validate,
   1034 # check if columns specified as unique
   1035 # are in fact unique.
   1036 if validate is not None:
-> 1037     self._validate_validate_kwd(validate)

File ~/work/pandas/pandas/pandas/core/reshape/merge.py:2005, in _MergeOperation._va
   2000         raise MergeError(
   2001             "Merge keys are not unique in left dataset; not a one-to-one me
   2002             f"{left_error_msg(left_join_index)}"
   2003         )
   2004     if not right_unique:
-> 2005         raise MergeError(
   2006             "Merge keys are not unique in right dataset; not a one-to-one m
   2007             f"{right_error_msg(right_join_index)}"
   2008         )
   2010 elif validate in ["one_to_many", "1:m"]:
   2011     if not left_unique:

MergeError: Merge keys are not unique in right dataset; not a one-to-one merge
Duplicates in right:
 B
 2
 2 ...
```

If the user is aware of the duplicates in the right `DataFrame` but wants to ensure there are no duplicates in the left `DataFrame`, one can use the `validate='one_to_many'` argument instead, which will not raise an exception.

```
In [72]: pd.merge(left, right, on="B", how="outer", validate="one_to_many")
Out[72]: 
   A_x  B  A_y
0    1  1  NaN
1    2  2  4.0
2    2  2  5.0
3    2  2  6.0
```

## Merge result indicator

`merge()` accepts the argument `indicator`. If `True`, a Categorical-type column called `_merge` will be added to the output object that takes on values:

| Observation Origin | `_merge` value |
|---|---|
| Merge key only in `'left'` frame | `left_only` |
| Merge key only in `'right'` frame | `right_only` |
| Merge key in both frames | `both` |

```
In [73]: df1 = pd.DataFrame({"col1": [0, 1], "col_left": ["a", "b"]})

In [74]: df2 = pd.DataFrame({"col1": [1, 2, 2], "col_right": [2, 2, 2]})

In [75]: pd.merge(df1, df2, on="col1", how="outer", indicator=True)
Out[75]: 
   col1 col_left  col_right      _merge
0     0        a        NaN   left_only
1     1        b        2.0        both
2     2      NaN        2.0  right_only
3     2      NaN        2.0  right_only
```

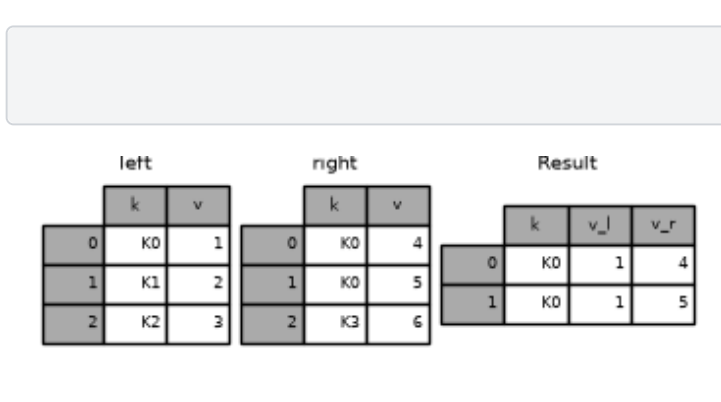
A string argument to `indicator` will use the value as the name for the indicator column.

```
In [76]: pd.merge(df1, df2, on="col1", how="outer", indicator="indicator_column")
Out[76]: 
   col1 col_left  col_right indicator_column
0     0        a        NaN        left_only
1     1        b        2.0             both
2     2      NaN        2.0       right_only
3     2      NaN        2.0       right_only
```

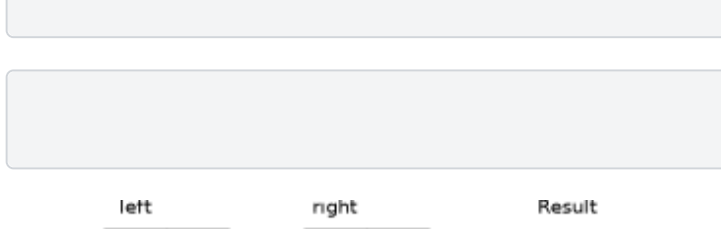
## Overlapping value columns

The merge `suffixes` argument takes a tuple or list of strings to append to overlapping column names in the input `DataFrame` to disambiguate the result columns:

```
In [77]: left = pd.DataFrame({"k": ["K0", "K1", "K2"], "v": [1, 2, 3]})

In [78]: right = pd.DataFrame({"k": ["K0", "K0", "K3"], "v": [4, 5, 6]})

In [79]: result = pd.merge(left, right, on="k")

In [80]: result
Out[80]: 
    k  v_x  v_y
0  K0    1    4
1  K0    1    5
```



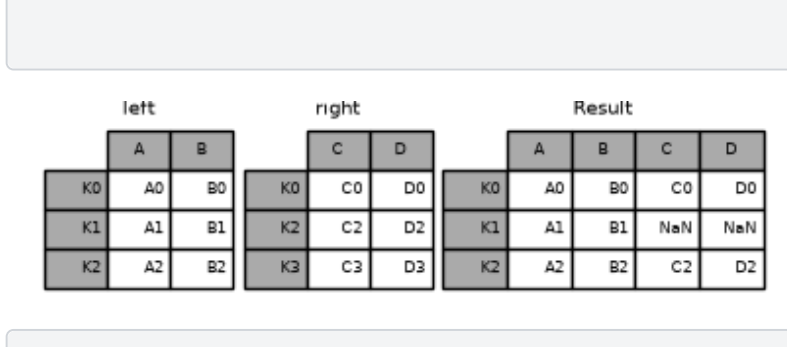
```
In [81]: result = pd.merge(left, right, on="k", suffixes=("_l", "_r"))

In [82]: result
Out[82]: 
    k  v_l  v_r
0  K0    1    4
1  K0    1    5
```



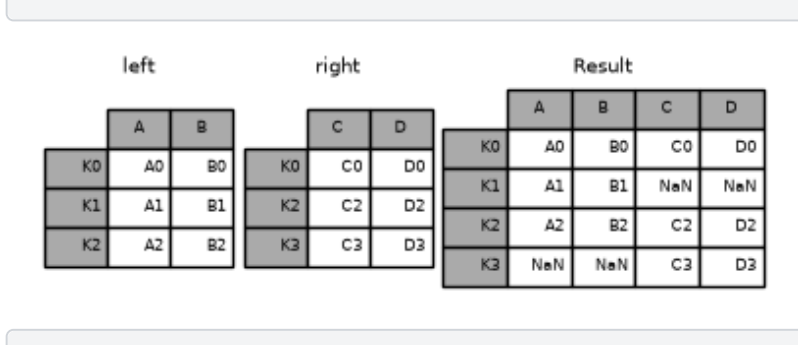
## `DataFrame.join()`

`DataFrame.join()` combines the columns of multiple, potentially differently-indexed `DataFrame` into a single result `DataFrame`.

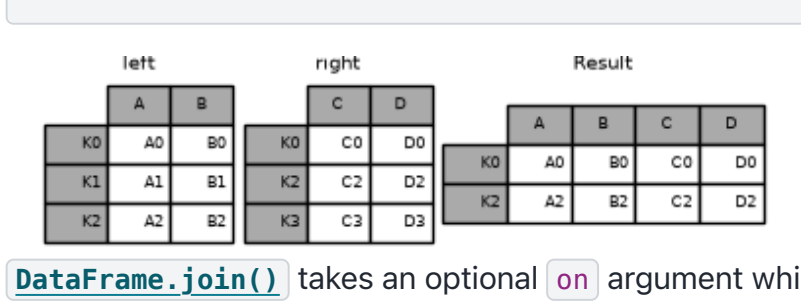
```
In [83]: left = pd.DataFrame(
   ....:     {"A": ["A0", "A1", "A2"], "B": ["B0", "B1", "B2"]}, index=["K0", "K1",
   ....: )
   ....: 

In [84]: right = pd.DataFrame(
   ....:     {"C": ["C0", "C2", "C3"], "D": ["D0", "D2", "D3"]}, index=["K0", "K2",
   ....: )
   ....: 

In [85]: result = left.join(right)

In [86]: result
Out[86]: 
     A   B    C    D
K0  A0  B0   C0   D0
K1  A1  B1  NaN  NaN
K2  A2  B2   C2   D2
```



```
In [87]: result = left.join(right, how="outer")

In [88]: result
Out[88]: 
      A    B    C    D
K0   A0   B0   C0   D0
K1   A1   B1  NaN  NaN
K2   A2   B2   C2   D2
K3  NaN  NaN   C3   D3
```

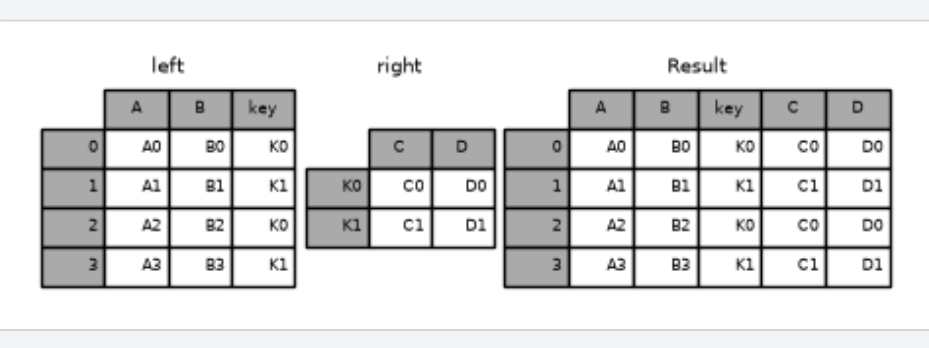


```
In [89]: result = left.join(right, how="inner")

In [90]: result
Out[90]: 
     A   B   C   D
K0  A0  B0  C0  D0
K2  A2  B2  C2  D2
```

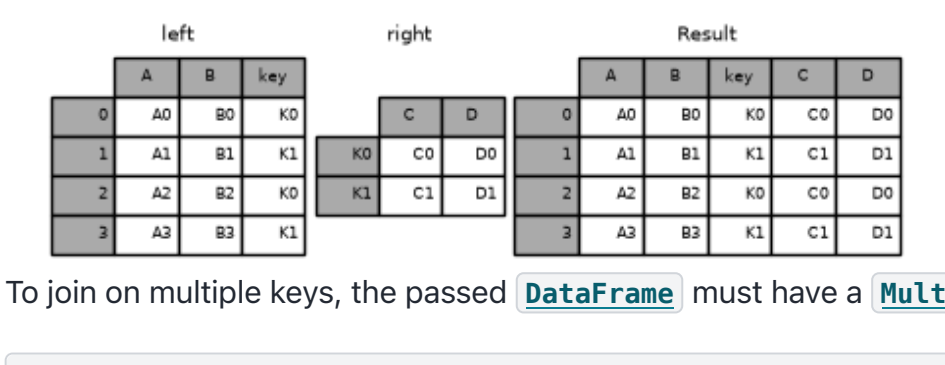


`DataFrame.join()` takes an optional `on` argument which may be a column or multiple column names that the passed `DataFrame` is to be aligned.

```
In [91]: left = pd.DataFrame(
   ....:     {
   ....:         "A": ["A0", "A1", "A2", "A3"],
   ....:         "B": ["B0", "B1", "B2", "B3"],
   ....:         "key": ["K0", "K1", "K0", "K1"],
   ....:     }
   ....: )
   ....: 

In [92]: right = pd.DataFrame({"C": ["C0", "C1"], "D": ["D0", "D1"]}, index=["K0", "

In [93]: result = left.join(right, on="key")

In [94]: result
Out[94]: 
    A   B key   C   D
0  A0  B0  K0  C0  D0
1  A1  B1  K1  C1  D1
2  A2  B2  K0  C0  D0
3  A3  B3  K1  C1  D1
```

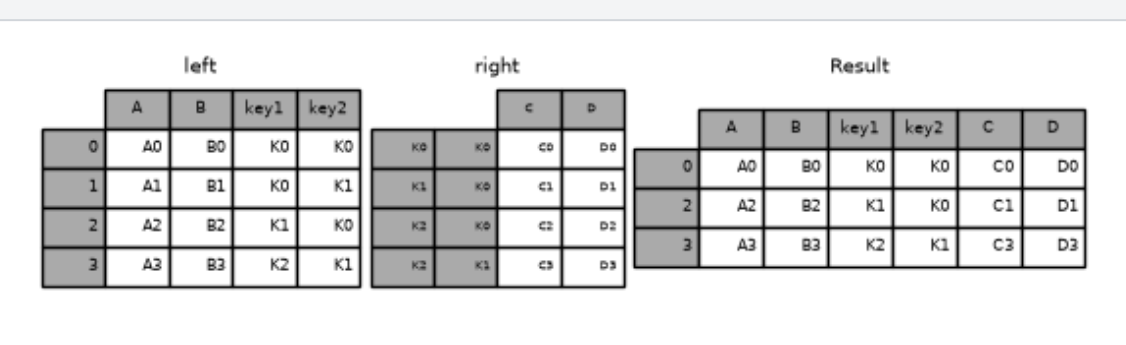


```
In [95]: result = pd.merge(
   ....:     left, right, left_on="key", right_index=True, how="left", sort=False
   ....: )
   ....: 

In [96]: result
Out[96]: 
    A   B key   C   D
0  A0  B0  K0  C0  D0
1  A1  B1  K1  C1  D1
2  A2  B2  K0  C0  D0
3  A3  B3  K1  C1  D1
```



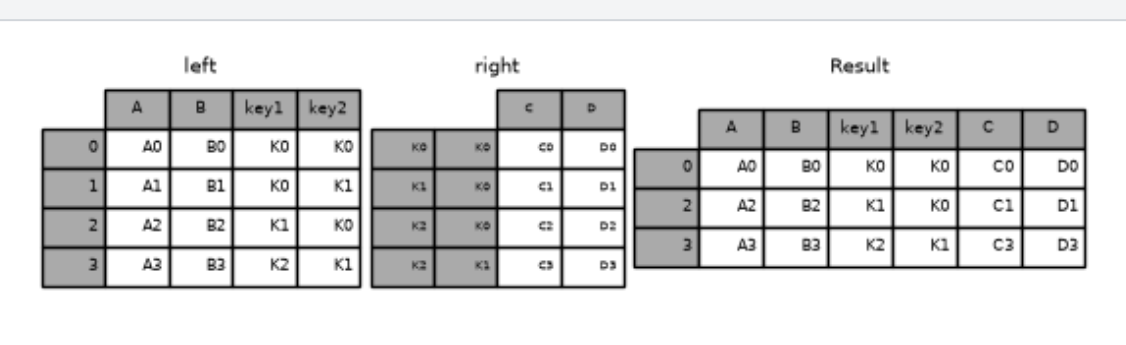
To join on multiple keys, the passed `DataFrame` must have a `MultiIndex`:

```
In [97]: left = pd.DataFrame(
   ....:     {
   ....:         "A": ["A0", "A1", "A2", "A3"],
   ....:         "B": ["B0", "B1", "B2", "B3"],
   ....:         "key1": ["K0", "K0", "K1", "K2"],
   ....:         "key2": ["K0", "K1", "K0", "K1"],
   ....:     }
   ....: )
   ....: 

In [98]: index = pd.MultiIndex.from_tuples(
   ....:     [("K0", "K0"), ("K1", "K0"), ("K2", "K0"), ("K2", "K1")]
   ....: )
   ....: 

In [99]: right = pd.DataFrame(
   ....:     {"C": ["C0", "C1", "C2", "C3"], "D": ["D0", "D1", "D2", "D3"]}, index=
   ....: )
   ....: 

In [100]: result = left.join(right, on=["key1", "key2"])

In [101]: result
Out[101]: 
    A   B key1 key2    C    D
0  A0  B0   K0   K0   C0   D0
1  A1  B1   K0   K1  NaN  NaN
2  A2  B2   K1   K0   C1   D1
3  A3  B3   K2   K1   C3   D3
```



The default for `DataFrame.join` is to perform a left join which uses only the keys found in the calling `DataFrame`. Other join types can be specified with `how`.

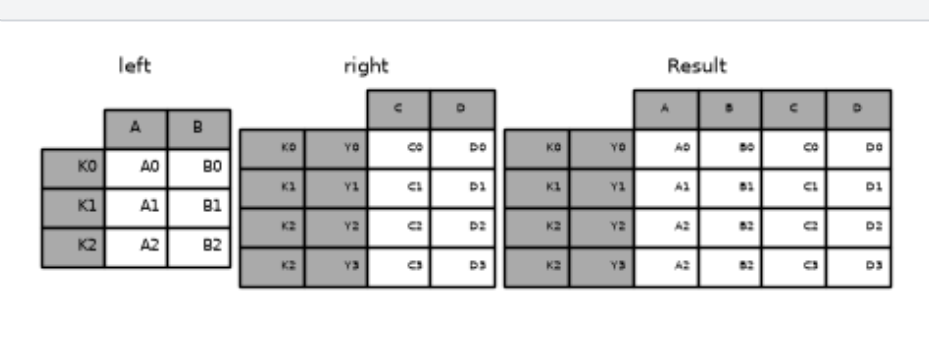
```
In [102]: result = left.join(right, on=["key1", "key2"], how="inner")

In [103]: result
Out[103]: 
    A   B key1 key2   C   D
0  A0  B0   K0   K0  C0  D0
2  A2  B2   K1   K0  C1  D1
3  A3  B3   K2   K1  C3  D3
```



## Joining a single Index to a MultiIndex

You can join a `DataFrame` with a `Index` to a `DataFrame` with a `MultiIndex` on a level. The `name` of the `Index` will match the level name of the `MultiIndex`.

```
In [104]: left = pd.DataFrame(
   .....:     {"A": ["A0", "A1", "A2"], "B": ["B0", "B1", "B2"]},
   .....:     index=pd.Index(["K0", "K1", "K2"], name="key"),
   .....: )
   .....: 

In [105]: index = pd.MultiIndex.from_tuples(
   .....:     [("K0", "Y0"), ("K1", "Y1"), ("K2", "Y2"), ("K2", "Y3")],
   .....:     names=["key", "Y"],
   .....: )
   .....: 

In [106]: right = pd.DataFrame(
   .....:     {"C": ["C0", "C1", "C2", "C3"], "D": ["D0", "D1", "D2", "D3"]},
   .....:     index=index,
   .....: )
   .....: 

In [107]: result = left.join(right, how="inner")

In [108]: result
Out[108]: 
         A   B   C   D
key Y                 
K0  Y0  A0  B0  C0  D0
K1  Y1  A1  B1  C1  D1
K2  Y2  A2  B2  C2  D2
    Y3  A2  B2  C3  D3
```

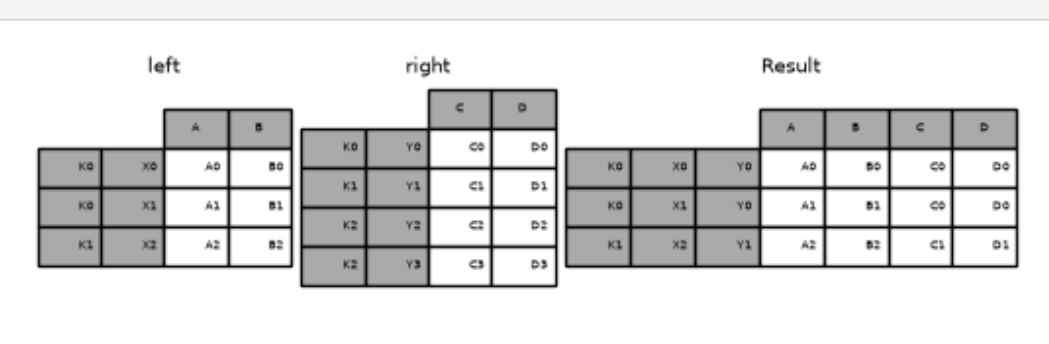


## Joining with two `MultiIndex`

The `MultiIndex` of the input argument must be completely used in the join and is a subset of the indices in the left argument.

```
In [109]: leftindex = pd.MultiIndex.from_product(
   .....:     [list("abc"), list("xy"), [1, 2]], names=["abc", "xy", "num"]
   .....: )
   .....: 

In [110]: left = pd.DataFrame({"v1": range(12)}, index=leftindex)

In [111]: left
Out[111]: 
            v1
abc xy num    
a   x  1     0
       2     1
    y  1     2
       2     3
b   x  1     4
       2     5
    y  1     6
       2     7
c   x  1     8
       2     9
    y  1    10
       2    11

In [112]: rightindex = pd.MultiIndex.from_product(
   .....:     [list("abc"), list("xy")], names=["abc", "xy"]
   .....: )
   .....: 

In [113]: right = pd.DataFrame({"v2": [100 * i for i in range(1, 7)]}, index=rightin

In [114]: right
Out[114]: 
         v2
abc xy     
a   x   100
    y   200
b   x   300
    y   400
c   x   500
    y   600

In [115]: left.join(right, on=["abc", "xy"], how="inner")
Out[115]: 
            v1   v2
abc xy num         
a   x  1     0  100
       2     1  100
    y  1     2  200
       2     3  200
b   x  1     4  300
       2     5  300
    y  1     6  400
       2     7  400
c   x  1     8  500
       2     9  500
    y  1    10  600
       2    11  600
```

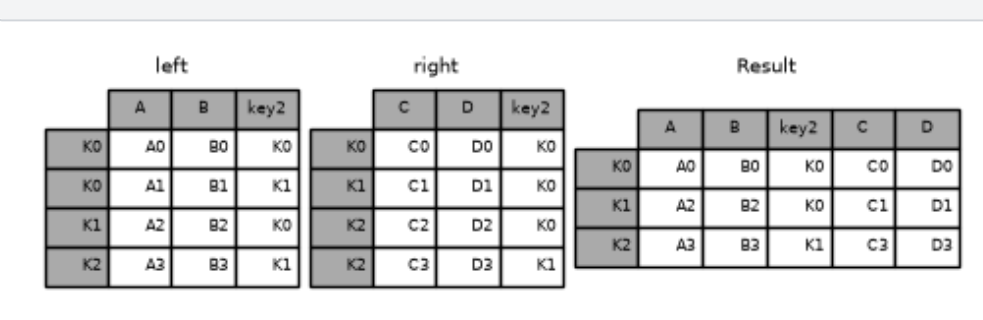
```
In [116]: leftindex = pd.MultiIndex.from_tuples(
   .....:     [("K0", "X0"), ("K0", "X1"), ("K1", "X2")], names=["key", "X"]
   .....: )
   .....: 

In [117]: left = pd.DataFrame(
   .....:     {"A": ["A0", "A1", "A2"], "B": ["B0", "B1", "B2"]}, index=leftindex
   .....: )
   .....: 

In [118]: rightindex = pd.MultiIndex.from_tuples(
   .....:     [("K0", "Y0"), ("K1", "Y1"), ("K2", "Y2"), ("K2", "Y3")], names=["key"
   .....: )
   .....: 

In [119]: right = pd.DataFrame(
   .....:     {"C": ["C0", "C1", "C2", "C3"], "D": ["D0", "D1", "D2", "D3"]}, index=
   .....: )
   .....: 

In [120]: result = pd.merge(
   .....:     left.reset_index(), right.reset_index(), on=["key"], how="inner"
   .....: ).set_index(["key", "X", "Y"])
   .....: 

In [121]: result
Out[121]: 
            A   B   C   D
key X  Y                 
K0  X0 Y0  A0  B0  C0  D0
    X1 Y0  A1  B1  C0  D0
K1  X2 Y1  A2  B2  C1  D1
```



## Merging on a combination of columns and index levels

Strings passed as the `on`, `left_on`, and `right_on` parameters may refer to either column names or index level names. This enables merging `DataFrame` instances on a combination of index levels and columns without resetting indexes.

```
In [122]: left_index = pd.Index(["K0", "K0", "K1", "K2"], name="key1")

In [123]: left = pd.DataFrame(
   .....:     {
   .....:         "A": ["A0", "A1", "A2", "A3"],
   .....:         "B": ["B0", "B1", "B2", "B3"],
   .....:         "key2": ["K0", "K1", "K0", "K1"],
   .....:     },
   .....:     index=left_index,
   .....: )
   .....: 

In [124]: right_index = pd.Index(["K0", "K1", "K2", "K2"], name="key1")

In [125]: right = pd.DataFrame(
   .....:     {
   .....:         "C": ["C0", "C1", "C2", "C3"],
   .....:         "D": ["D0", "D1", "D2", "D3"],
   .....:         "key2": ["K0", "K0", "K0", "K1"],
   .....:     },
   .....:     index=right_index,
   .....: )
   .....: 

In [126]: result = left.merge(right, on=["key1", "key2"])

In [127]: result
Out[127]: 
       A   B key2   C   D
key1                     
K0    A0  B0   K0  C0  D0
K1    A2  B2   K0  C1  D1
K2    A3  B3   K1  C3  D3
```



**Note**

When `DataFrame` are joined on a string that matches an index level in both arguments, the index level is preserved as an index level in the resulting `DataFrame`.

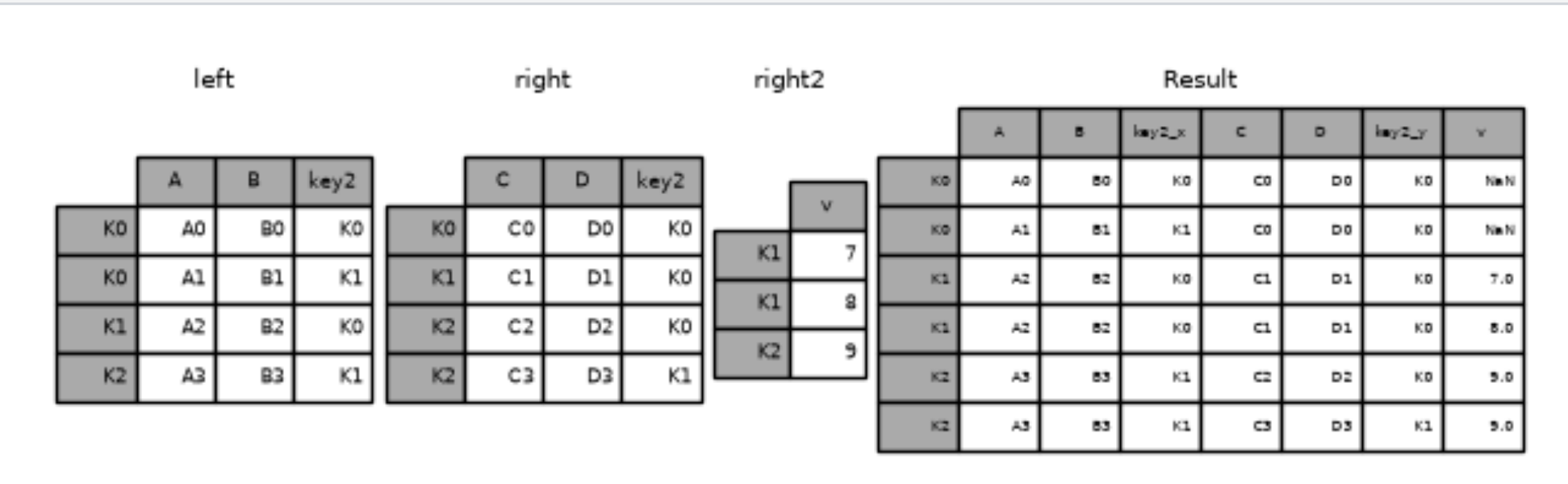
**Note**

When `DataFrame` are joined using only some of the levels of a `MultiIndex`, the extra levels will be dropped from the resulting join. To preserve those levels, use `DataFrame.reset_index()` on those level names to move those levels to columns prior to the join.

## Joining multiple `DataFrame`

A list or tuple of `DataFrame` can also be passed to `join()` to join them together on their indexes.

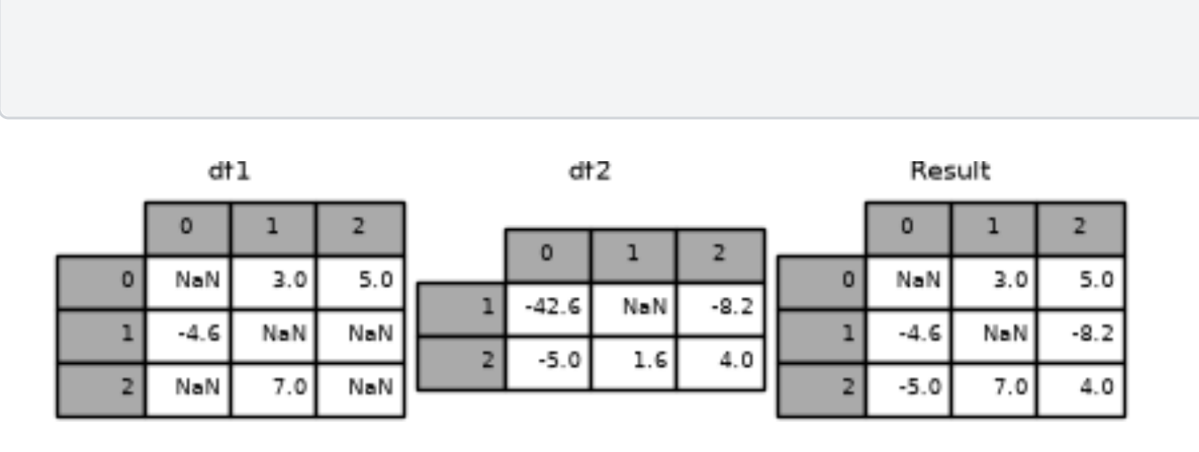
```
In [128]: right2 = pd.DataFrame({"v": [7, 8, 9]}, index=["K1", "K1", "K2"])

In [129]: result = left.join([right, right2])
```



## `DataFrame.combine_first()`

`DataFrame.combine_first()` update missing values from one `DataFrame` with the non-missing values in another `DataFrame` in the corresponding location.

```
In [130]: df1 = pd.DataFrame(
   .....:     [[np.nan, 3.0, 5.0], [-4.6, np.nan, np.nan], [np.nan, 7.0, np.nan]]
   .....: )
   .....: 

In [131]: df2 = pd.DataFrame([[-42.6, np.nan, -8.2], [-5.0, 1.6, 4]], index=[1, 2])

In [132]: result = df1.combine_first(df2)

In [133]: result
Out[133]: 
     0    1    2
0  NaN  3.0  5.0
1 -4.6  NaN -8.2
2 -5.0  7.0  4.0
```



## `merge_ordered()`

`merge_ordered()` combines ordered data such as numeric or time series data with optional filling of missing data with `fill_method`.

```
In [134]: left = pd.DataFrame(
   .....:     {"k": ["K0", "K1", "K1", "K2"], "lv": [1, 2, 3, 4], "s": ["a", "b", "
   .....: )
   .....: 

In [135]: right = pd.DataFrame({"k": ["K1", "K2", "K4"], "rv": [1, 2, 3]})

In [136]: pd.merge_ordered(left, right, fill_method="ffill", left_by="s")
Out[136]: 
     k   lv  s   rv
0   K0  1.0  a  NaN
1   K1  1.0  a  1.0
2   K2  1.0  a  2.0
3   K4  1.0  a  3.0
4   K1  2.0  b  1.0
5   K2  2.0  b  2.0
6   K4  2.0  b  3.0
7   K1  3.0  c  1.0
8   K2  3.0  c  2.0
9   K4  3.0  c  3.0
10  K1  NaN  d  1.0
11  K2  4.0  d  2.0
12  K4  4.0  d  3.0
```

## `merge_asof()`

`merge_asof()` is similar to an ordered left-join except that matches are on the nearest key rather than equal keys. For each row in the `left` `DataFrame`, the last row in the `right` `DataFrame` is selected where the `on` key is less than the left's key. Both `DataFrame` must be sorted by the key.

Optionally `merge_asof()` can perform a group-wise merge by matching the `by` key in addition to the nearest match on the `on` key.

```
In [137]: trades = pd.DataFrame(
   .....:     {
   .....:         "time": pd.to_datetime(
   .....:             [
   .....:                 "20160525 13:30:00.023",
   .....:                 "20160525 13:30:00.038",
   .....:                 "20160525 13:30:00.048",
   .....:                 "20160525 13:30:00.048",
   .....:                 "20160525 13:30:00.048",
   .....:             ]
   .....:         ),
   .....:         "ticker": ["MSFT", "MSFT", "GOOG", "GOOG", "AAPL"],
   .....:         "price": [51.95, 51.95, 720.77, 720.92, 98.00],
   .....:         "quantity": [75, 155, 100, 100, 100],
   .....:     },
   .....:     columns=["time", "ticker", "price", "quantity"],
   .....: )
   .....: 

In [138]: quotes = pd.DataFrame(
   .....:     {
   .....:         "time": pd.to_datetime(
   .....:             [
   .....:                 "20160525 13:30:00.023",
   .....:                 "20160525 13:30:00.023",
   .....:                 "20160525 13:30:00.030",
   .....:                 "20160525 13:30:00.041",
   .....:                 "20160525 13:30:00.048",
   .....:                 "20160525 13:30:00.049",
   .....:                 "20160525 13:30:00.072",
   .....:                 "20160525 13:30:00.075",
   .....:             ]
   .....:         ),
   .....:         "ticker": ["GOOG", "MSFT", "MSFT", "MSFT", "GOOG", "AAPL", "GOOG"
   .....:         "bid": [720.50, 51.95, 51.97, 51.99, 720.50, 97.99, 720.50, 52.01
   .....:         "ask": [720.93, 51.96, 51.98, 52.00, 720.93, 98.01, 720.88, 52.03
   .....:     },
   .....:     columns=["time", "ticker", "bid", "ask"],
   .....: )
   .....: 

In [139]: trades
Out[139]: 
                     time ticker   price  quantity
0 2016-05-25 13:30:00.023   MSFT   51.95        75
1 2016-05-25 13:30:00.038   MSFT   51.95       155
2 2016-05-25 13:30:00.048   GOOG  720.77       100
3 2016-05-25 13:30:00.048   GOOG  720.92       100
4 2016-05-25 13:30:00.048   AAPL   98.00       100

In [140]: quotes
Out[140]: 
                     time ticker     bid     ask
0 2016-05-25 13:30:00.023   GOOG  720.50  720.93
1 2016-05-25 13:30:00.023   MSFT   51.95   51.96
2 2016-05-25 13:30:00.030   MSFT   51.97   51.98
3 2016-05-25 13:30:00.041   MSFT   51.99   52.00
4 2016-05-25 13:30:00.048   GOOG  720.50  720.93
5 2016-05-25 13:30:00.049   AAPL   97.99   98.01
6 2016-05-25 13:30:00.072   GOOG  720.50  720.88
7 2016-05-25 13:30:00.075   MSFT   52.01   52.03

In [141]: pd.merge_asof(trades, quotes, on="time", by="ticker")
Out[141]: 
                     time ticker   price  quantity     bid     ask
0 2016-05-25 13:30:00.023   MSFT   51.95        75   51.95   51.96
1 2016-05-25 13:30:00.038   MSFT   51.95       155   51.97   51.98
2 2016-05-25 13:30:00.048   GOOG  720.77       100  720.50  720.93
3 2016-05-25 13:30:00.048   GOOG  720.92       100  720.50  720.93
4 2016-05-25 13:30:00.048   AAPL   98.00       100     NaN     NaN
```

`merge_asof()` within `2ms` between the quote time and the trade time.

```
In [142]: pd.merge_asof(trades, quotes, on="time", by="ticker", tolerance=pd.Timede
Out[142]: 
                     time ticker   price  quantity     bid     ask
0 2016-05-25 13:30:00.023   MSFT   51.95        75   51.95   51.96
1 2016-05-25 13:30:00.038   MSFT   51.95       155     NaN     NaN
2 2016-05-25 13:30:00.048   GOOG  720.77       100  720.50  720.93
3 2016-05-25 13:30:00.048   GOOG  720.92       100  720.50  720.93
4 2016-05-25 13:30:00.048   AAPL   98.00       100     NaN     NaN
```

`merge_asof()` within `10ms` between the quote time and the trade time and exclude exact matches on time. Note that though we exclude the exact matches (of the quotes), prior quotes **do** propagate to that point in time.

```
In [143]: pd.merge_asof(
   .....:     trades,
   .....:     quotes,
   .....:     on="time",
   .....:     by="ticker",
   .....:     tolerance=pd.Timedelta("10ms"),
   .....:     allow_exact_matches=False,
   .....: )
   .....: 
Out[143]: 
                     time ticker   price  quantity    bid    ask
0 2016-05-25 13:30:00.023   MSFT   51.95        75    NaN    NaN
1 2016-05-25 13:30:00.038   MSFT   51.95       155  51.97  51.98
2 2016-05-25 13:30:00.048   GOOG  720.77       100    NaN    NaN
3 2016-05-25 13:30:00.048   GOOG  720.92       100    NaN    NaN
4 2016-05-25 13:30:00.048   AAPL   98.00       100    NaN    NaN
```

## `compare()`

The `Series.compare()` and `DataFrame.compare()` methods allow you to compare two `DataFrame` or `Series`, respectively, and summarize their differences.

```
In [144]: df = pd.DataFrame(
   .....:     {
   .....:         "col1": ["a", "a", "b", "b", "a"],
   .....:         "col2": [1.0, 2.0, 3.0, np.nan, 5.0],
   .....:         "col3": [1.0, 2.0, 3.0, 4.0, 5.0],
   .....:     },
   .....:     columns=["col1", "col2", "col3"],
   .....: )
   .....: 

In [145]: df
Out[145]: 
  col1  col2  col3
0    a   1.0   1.0
1    a   2.0   2.0
2    b   3.0   3.0
3    b   NaN   4.0
4    a   5.0   5.0

In [146]: df2 = df.copy()

In [147]: df2.loc[0, "col1"] = "c"

In [148]: df2.loc[2, "col3"] = 4.0

In [149]: df2
Out[149]: 
  col1  col2  col3
0    c   1.0   1.0
1    a   2.0   2.0
2    b   3.0   4.0
3    b   NaN   4.0
4    a   5.0   5.0

In [150]: df.compare(df2)
Out[150]: 
  col1       col3      
  self other self other
0    a     c  NaN   NaN
2  NaN   NaN  3.0   4.0
```

By default, if two corresponding values are equal, they will be shown as `NaN`. Furthermore, if all values in an entire row / column are equal, that row / column will be omitted from the result. The remaining differences will be aligned on columns.

Stack the differences on rows.

```
In [151]: df.compare(df2, align_axis=0)
Out[151]: 
        col1  col3
0 self     a   NaN
  other    c   NaN
2 self   NaN   3.0
  other  NaN   4.0
```

Keep all original rows and columns with `keep_shape=True`.

```
In [152]: df.compare(df2, keep_shape=True)
Out[152]: 
  col1       col2       col3      
  self other self other self other
0    a     c  NaN   NaN  NaN   NaN
1  NaN   NaN  NaN   NaN  NaN   NaN
2  NaN   NaN  NaN   NaN  3.0   4.0
3  NaN   NaN  NaN   NaN  NaN   NaN
4  NaN   NaN  NaN   NaN  NaN   NaN
```

Keep all the original values even if they are equal.

```
In [153]: df.compare(df2, keep_shape=True, keep_equal=True)
Out[153]: 
  col1       col2       col3      
  self other self other self other
0    a     c  1.0   1.0  1.0   1.0
1    a     a  2.0   2.0  2.0   2.0
2    b     b  3.0   3.0  3.0   4.0
3    b     b  NaN   NaN  4.0   4.0
4    a     a  5.0   5.0  5.0   5.0
```

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